Introduction to Engineering Design

Version 1. 0

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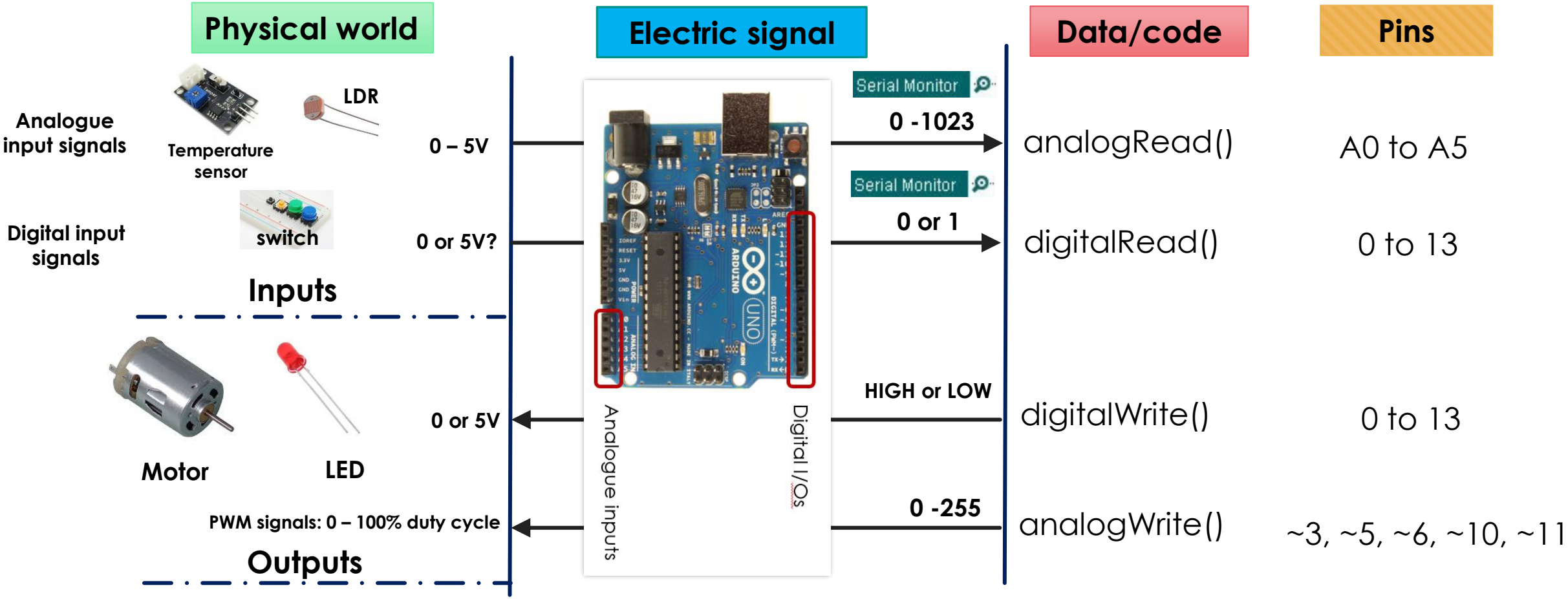
By Dr. Chia Chew Lin

The Hardware and the Interaction between the PINs and Arduino



Conceive

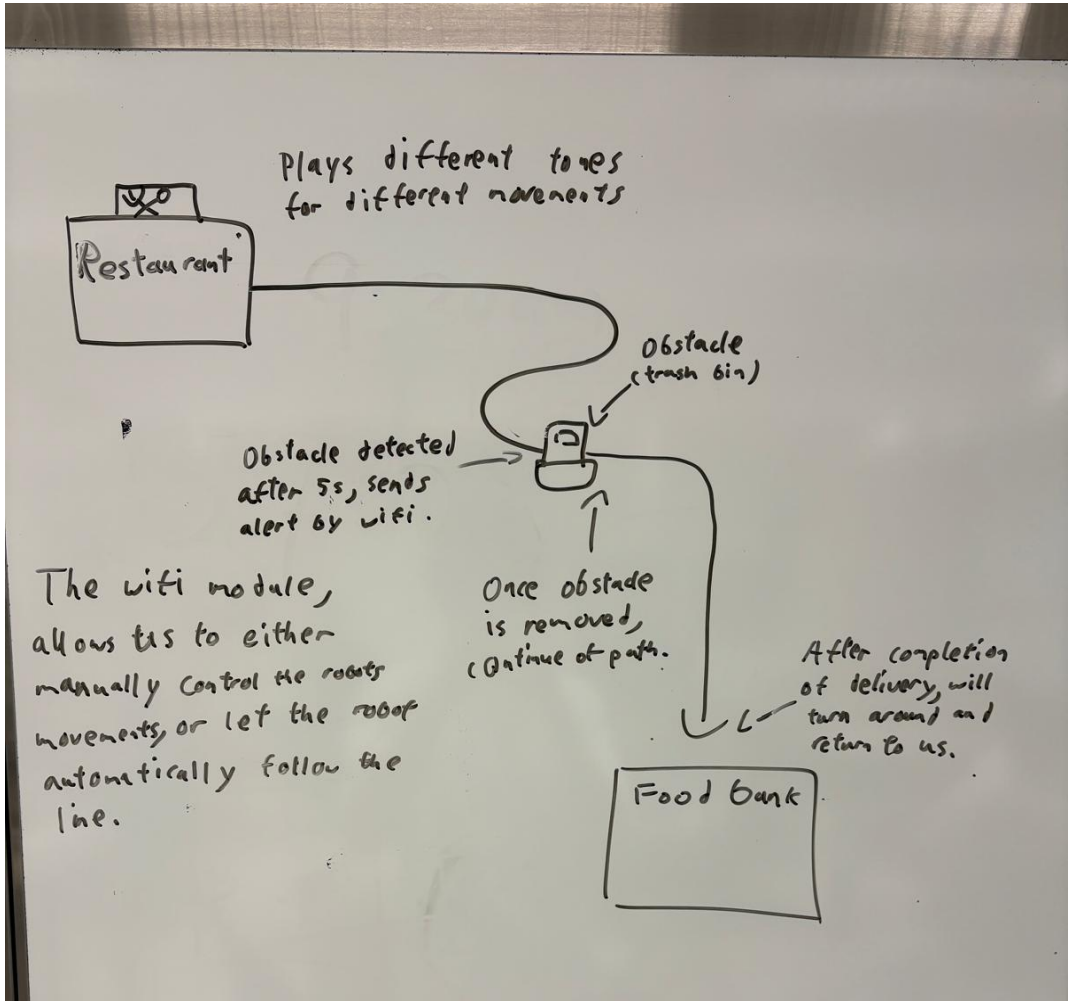
Problem Statement: How can we deliver unsold but edible food from supermarkets or restaurants to food banks or shelters.(UNSDG 2,3 and 12)

Solution: Develop embedded system with Arduino uno to transport food along a designated route, stopping for obstacles. After delivering the food, users can "scan" the robot's target board to allow it to turn back and return to its original location. Senders can do the same after it has returned, allowing them to send more food

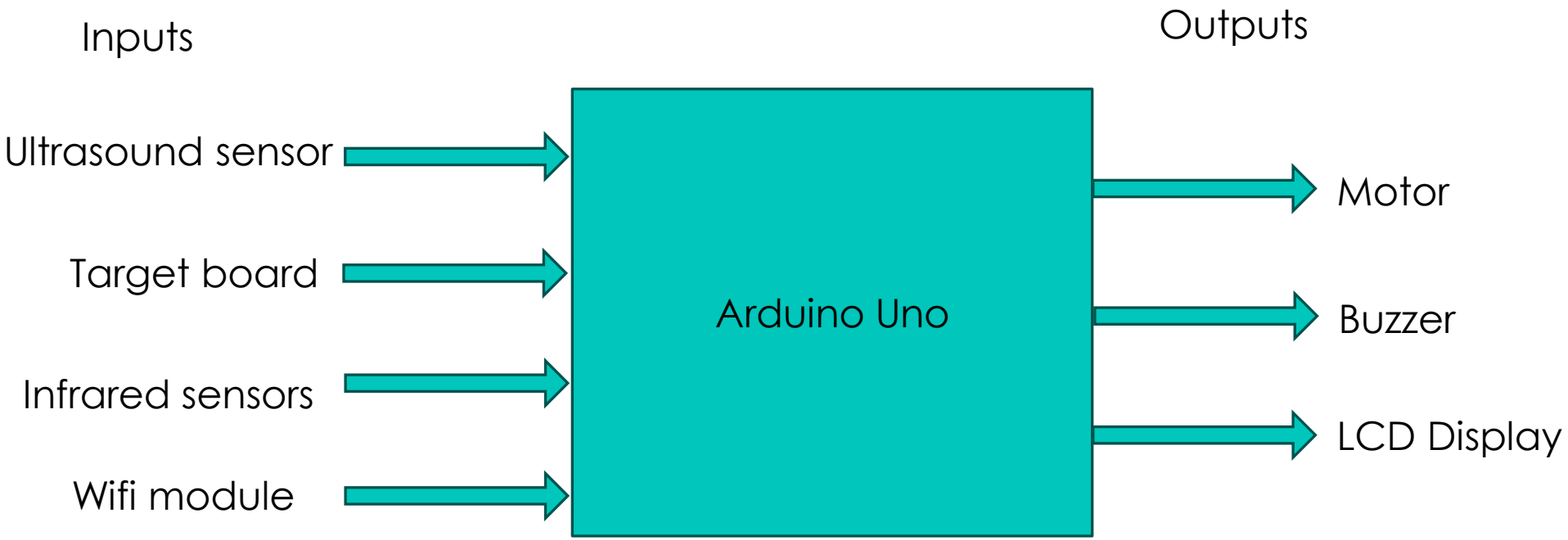
Sketch

During the process of the robot's automatic movement, the robot can run indefinitely. This allows the sender and receiver to use it without needing to reconfigure the robot. To do so, they can keep hitting the target board with light, allowing it to turn around and return.

In the event of an unforeseen accident like when the robot goes off-track, the sender can manually control the robot back to the designated route through the RemoteAccess app that uses the wifi module.



Design


Inputs
Outputs
Ultrasound sensor
Target board
Infrared sensors
Wifi module
Arduino Uno
Motor
Buzzer
LCD Display

Example: Overview of your program structure

Physical world

Write your Program based on your Zumo Strategy

Function call

handleInterrupt()

getdist()

drivemotor()

drivemotor()
processRoute()

- DRV8835 Dual Motor Driver

Movementsounds()

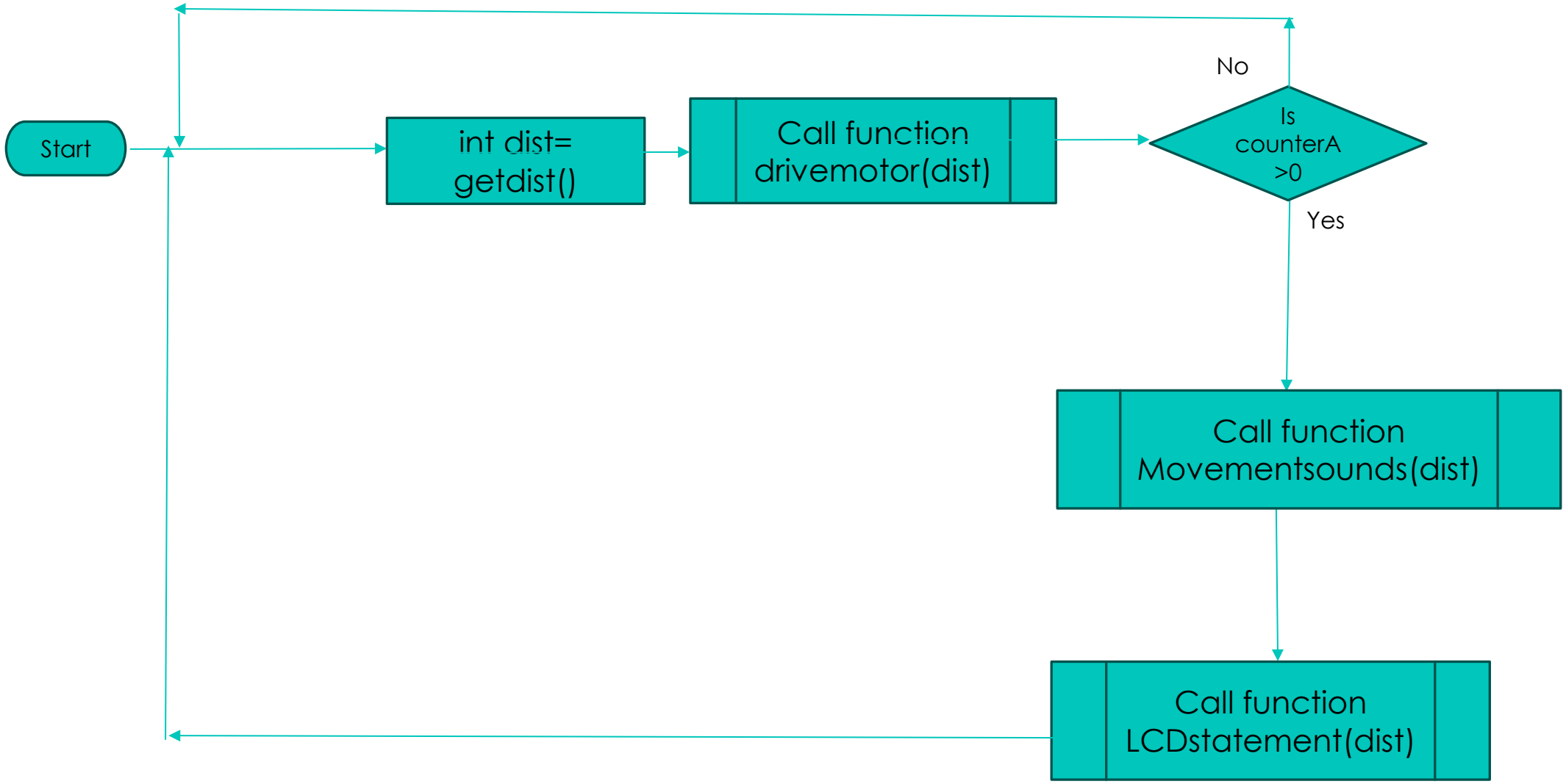
LCDstatement()
processRoute()

processRoute()

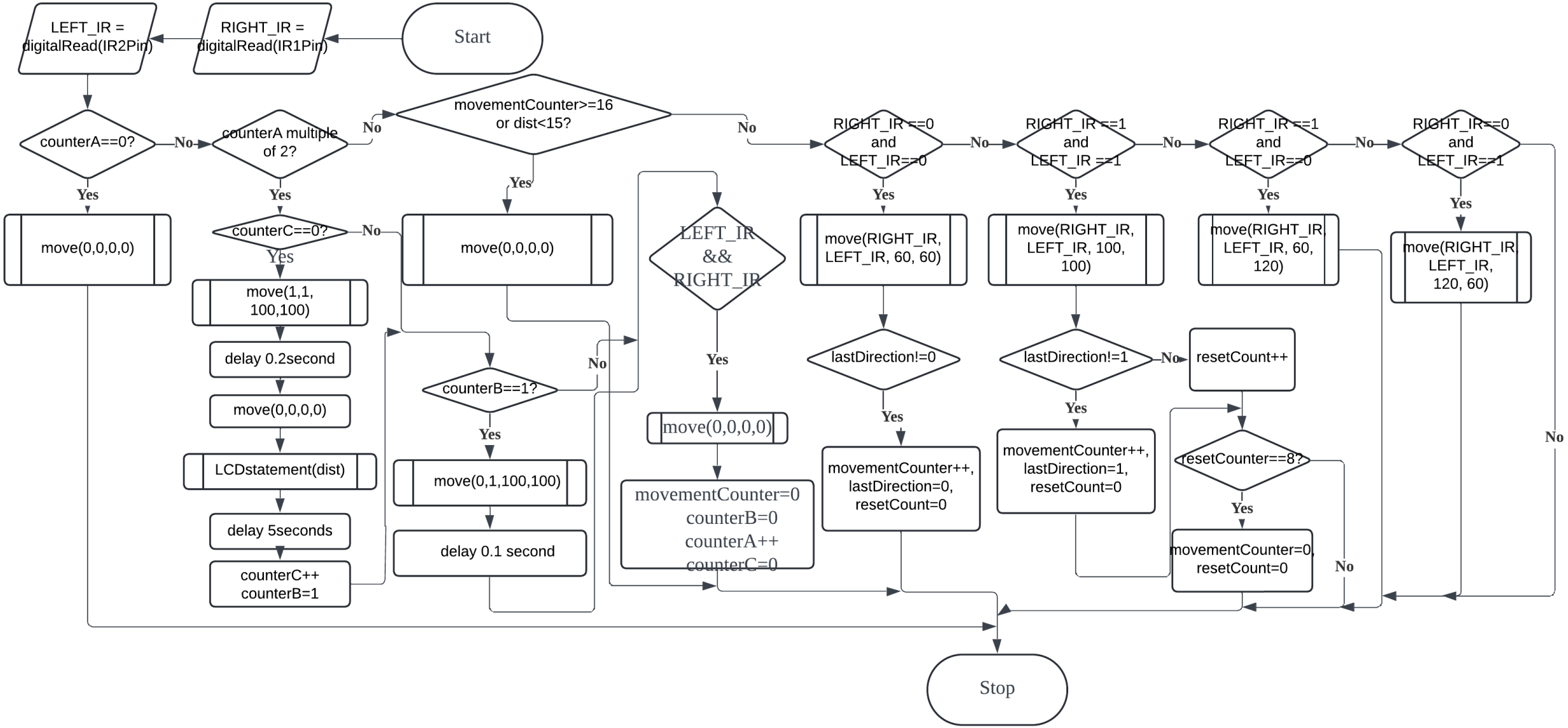
ESP LINK V1.0

- Make sure you have your basic functions written and working!
- You can have more functions

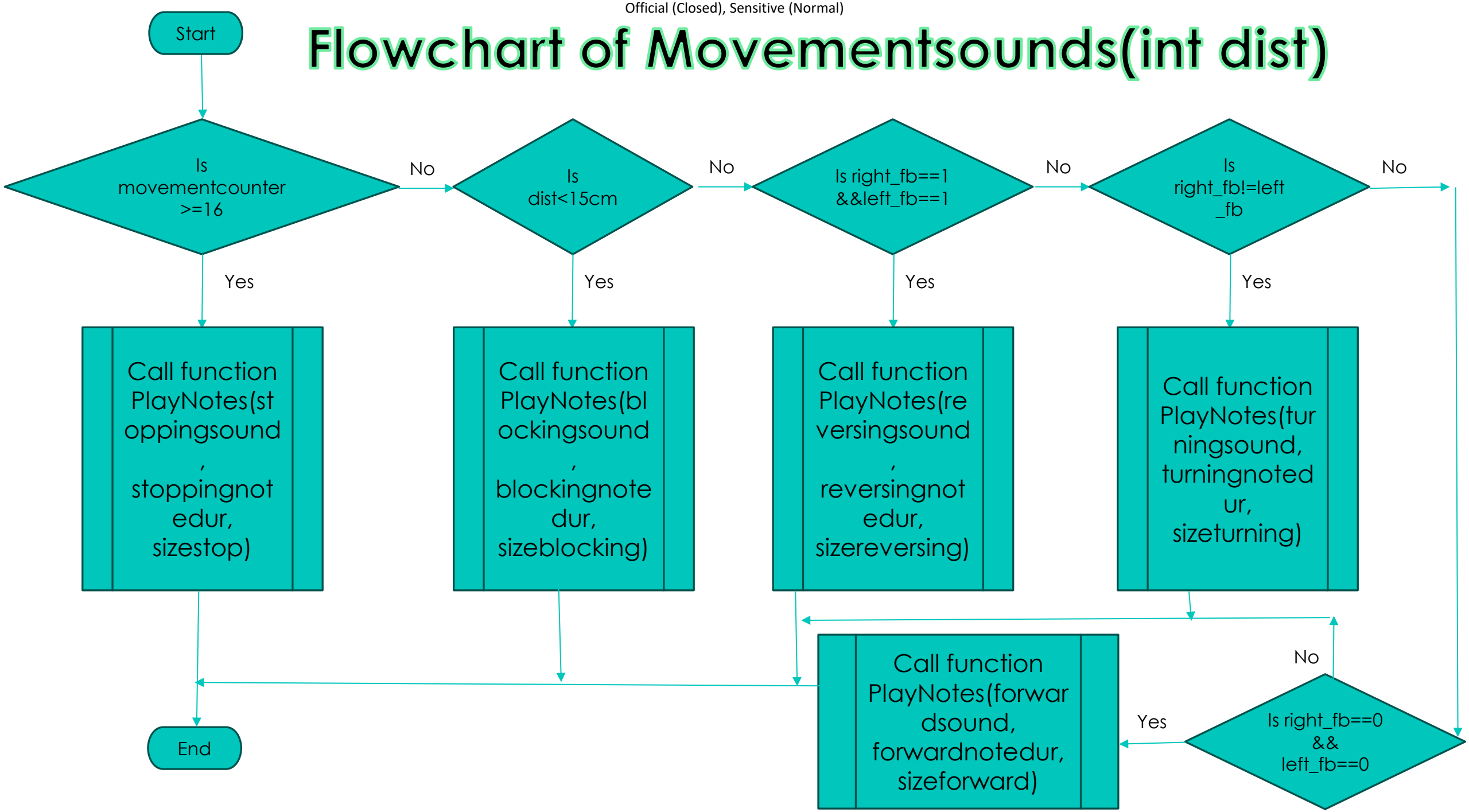
Flowchart of autoRun main loop()



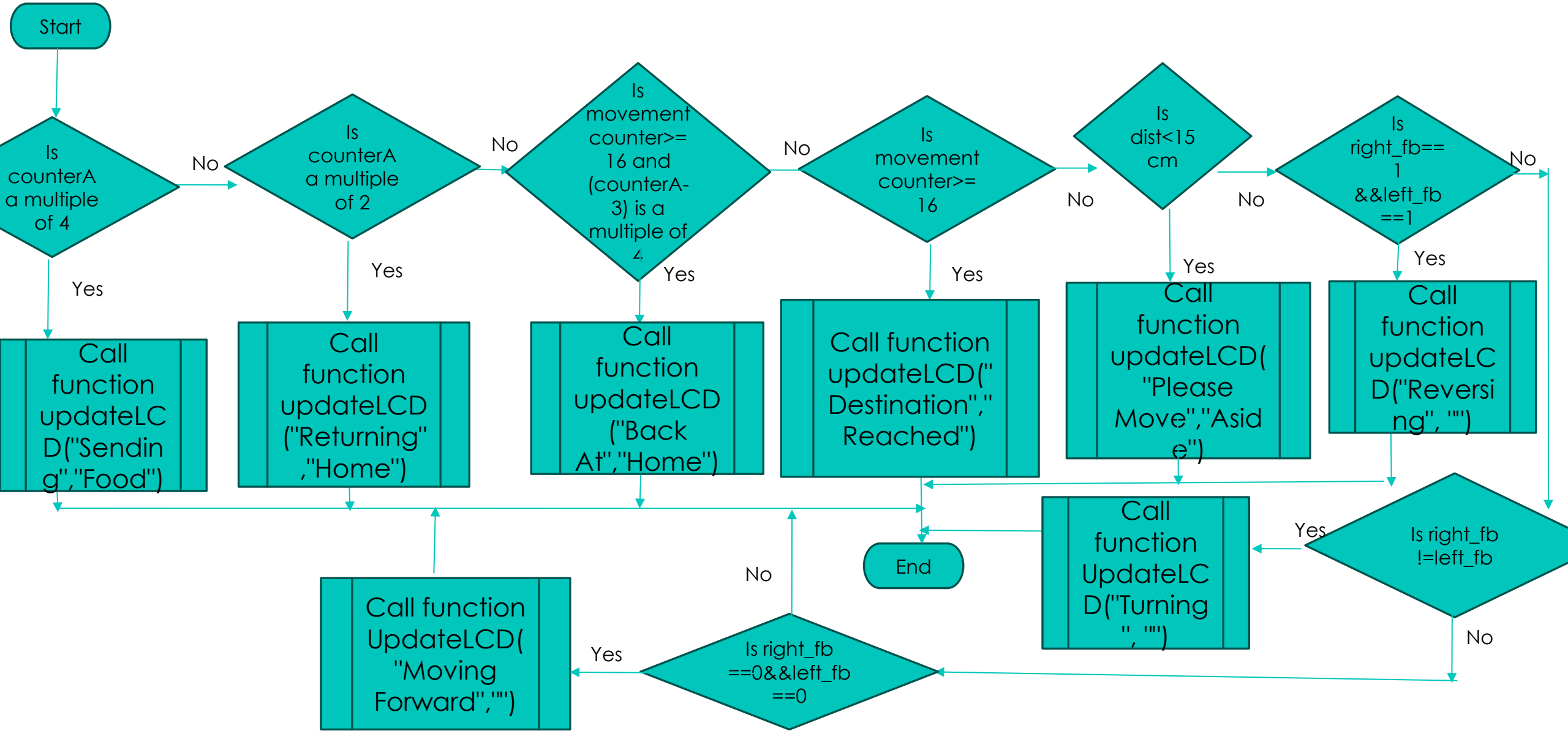
Flowchart of drivemotor(int dist)



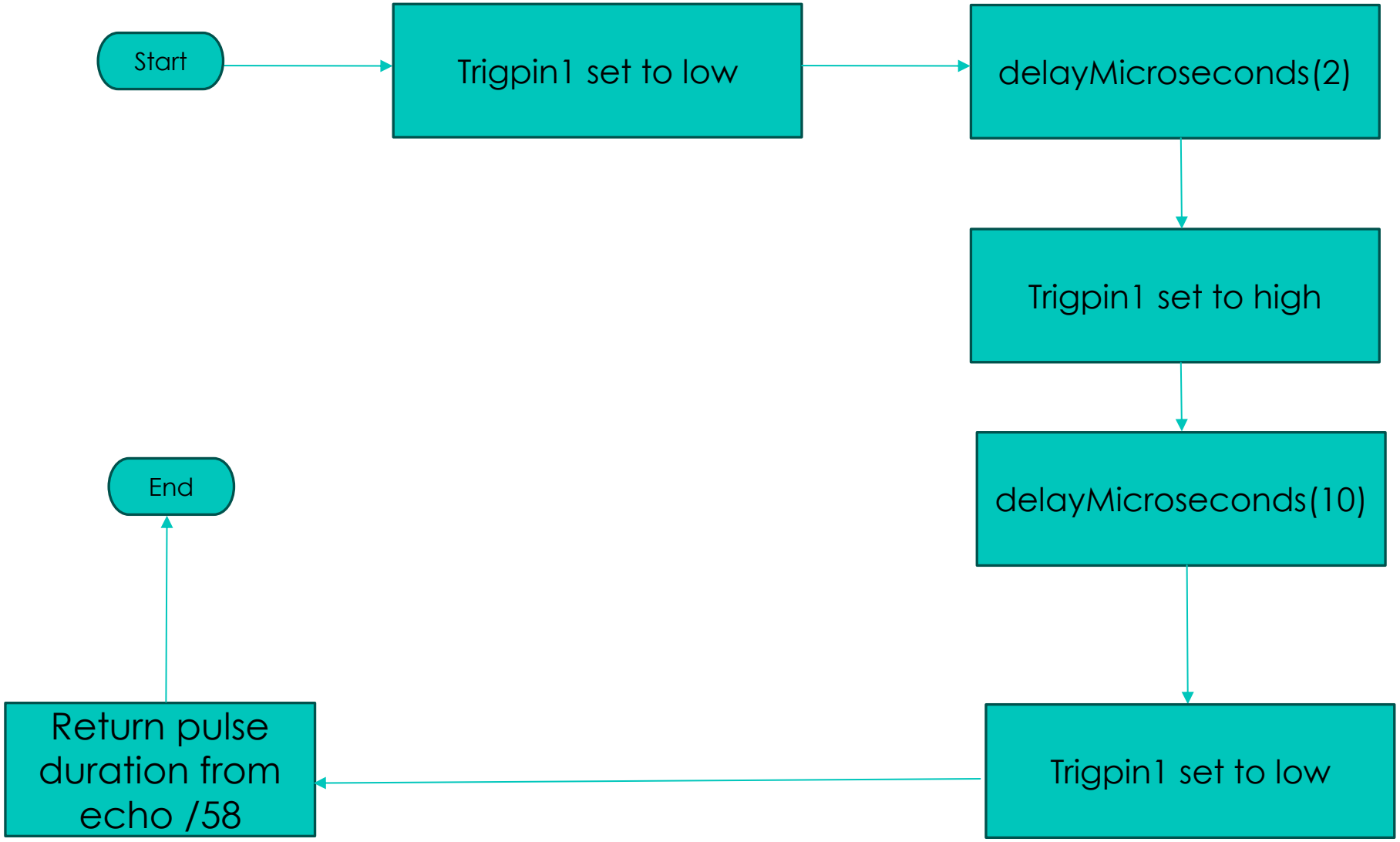
Flowchart of Movementsounds(int dist)



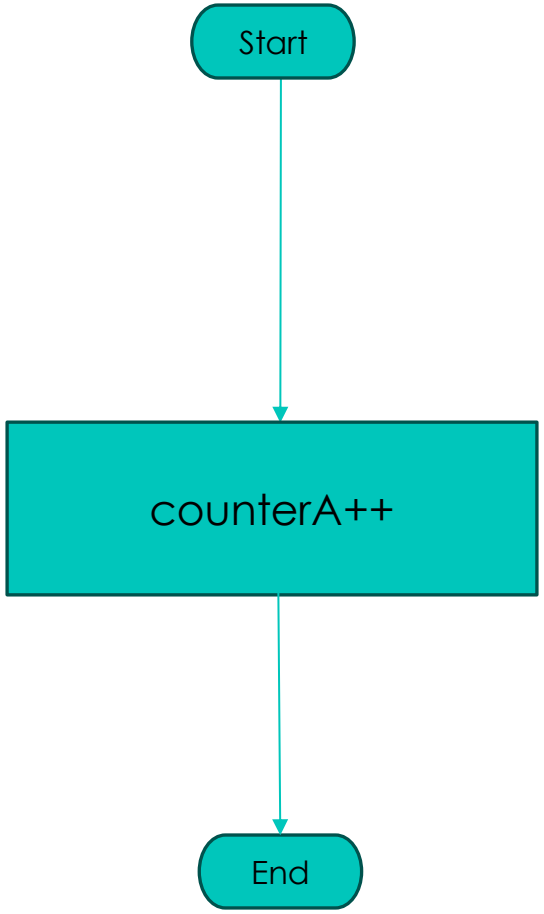
Flowchart of LCDstatement(int dist)



Flowchart of getdist()

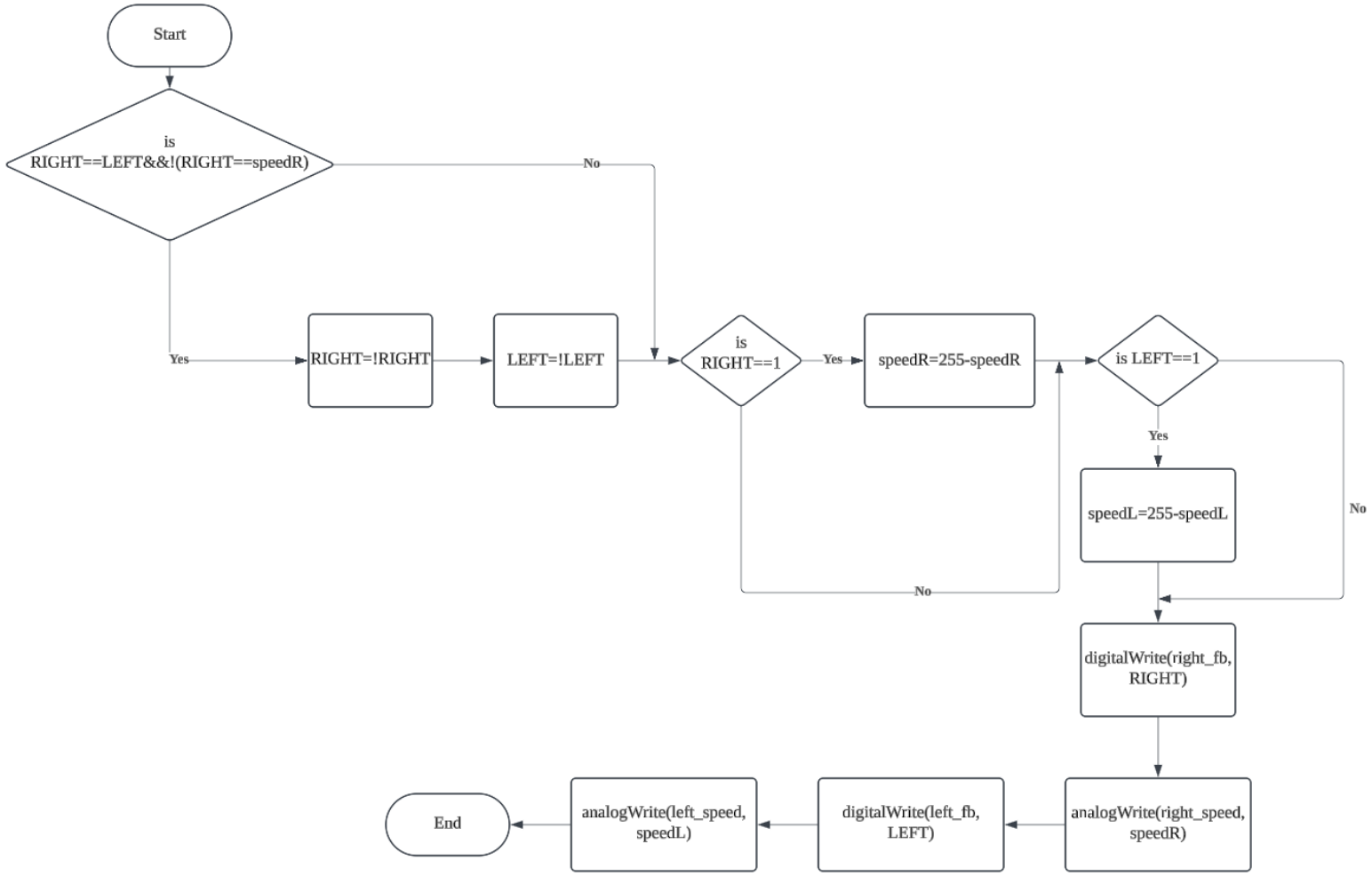


Flowchart of handleInterrupt()

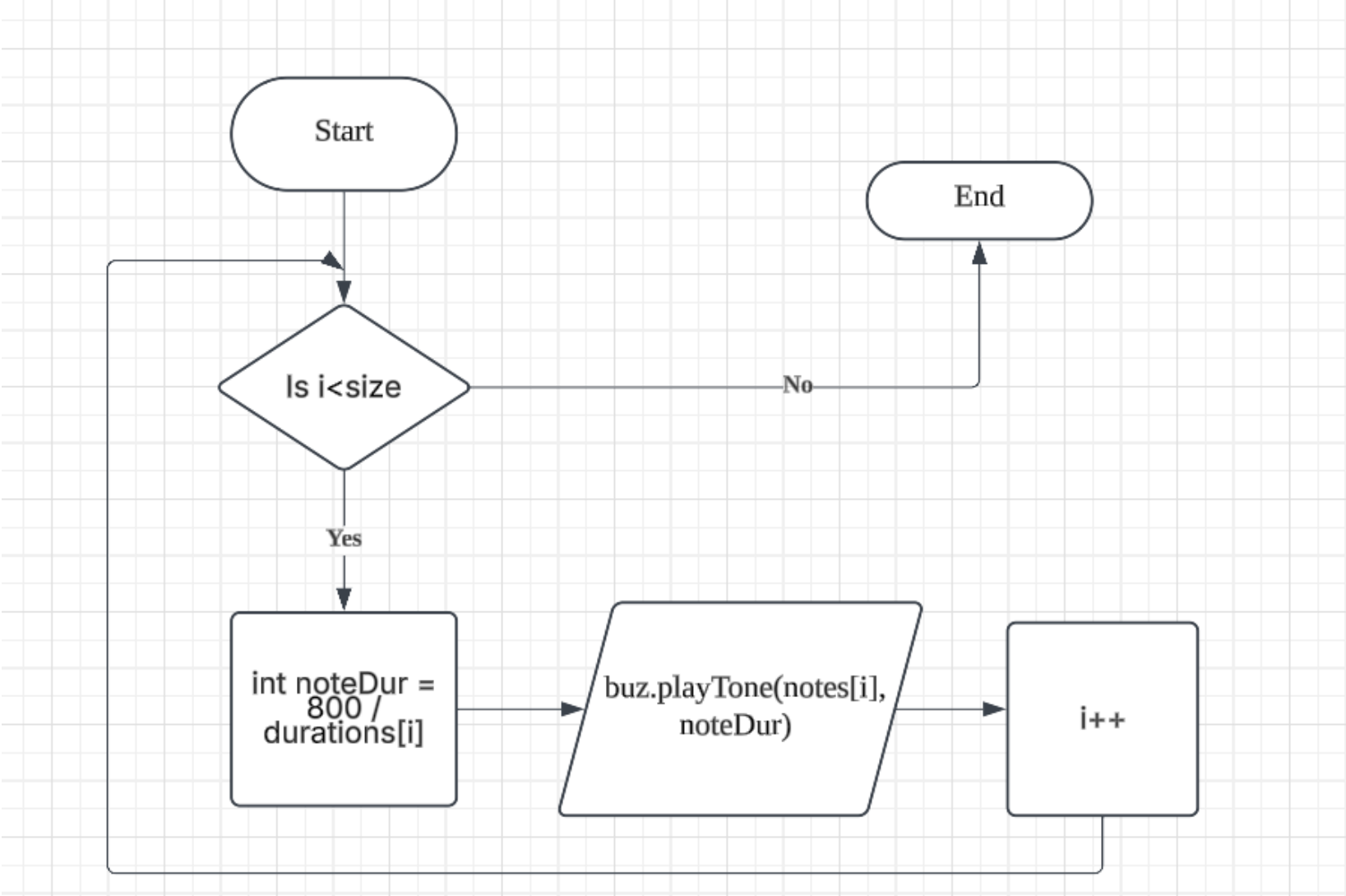


Flowchart of move(int RIGHT, int LEFT, int speedR, int speedL)

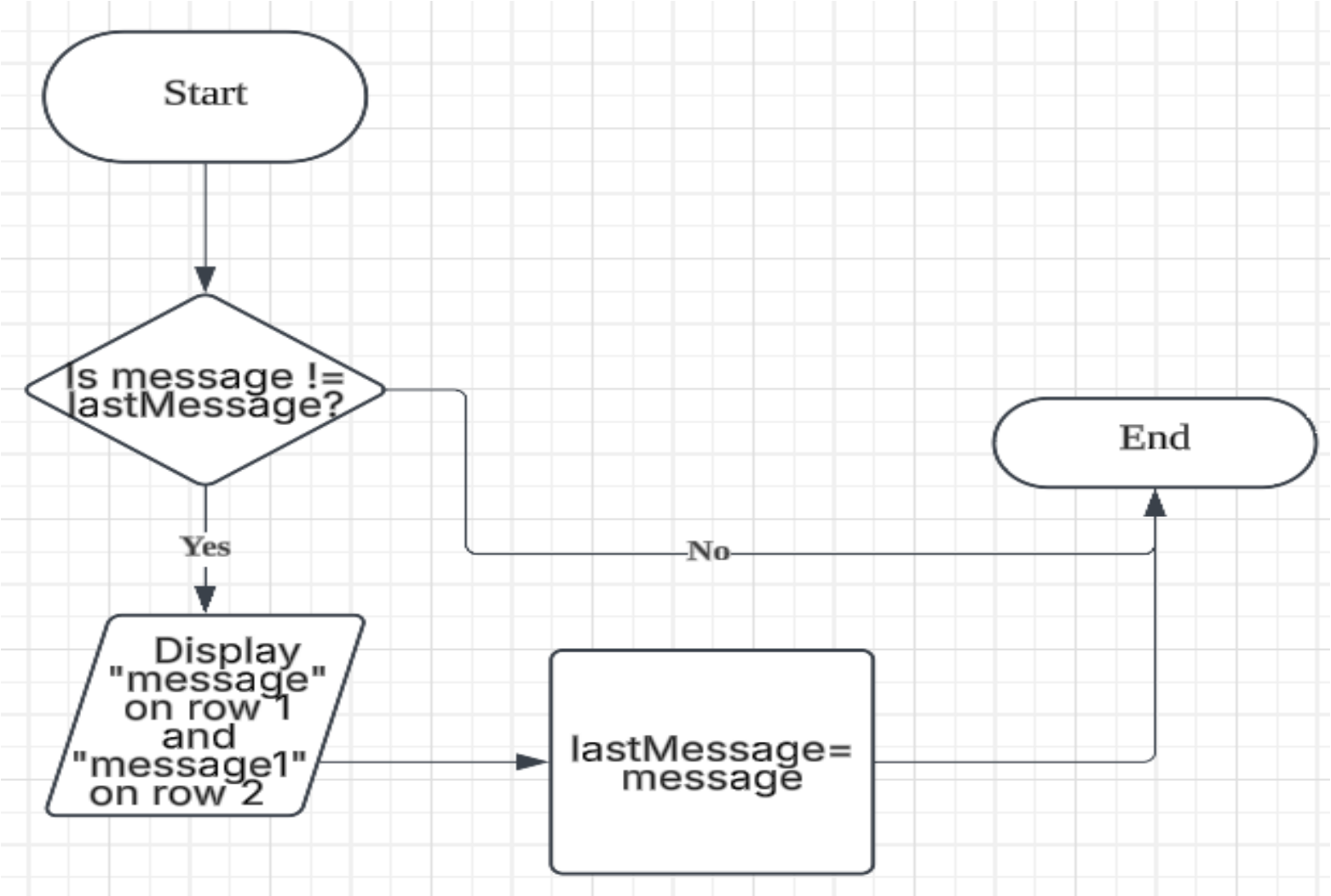
*This function is used for both autoRun and RemoteAccess



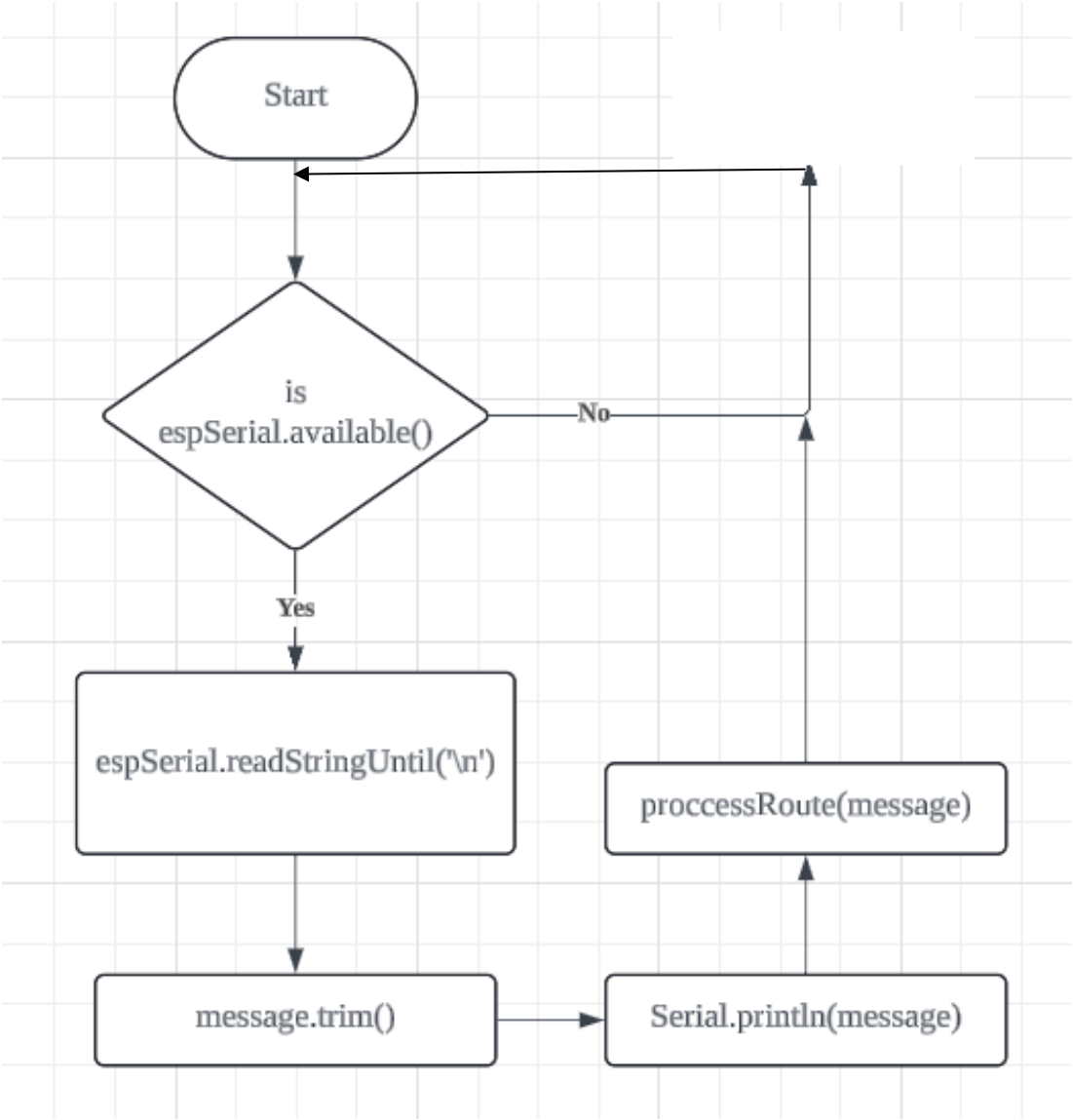
Flowchart of playNotes(const int *notes, const int *durations, int size)



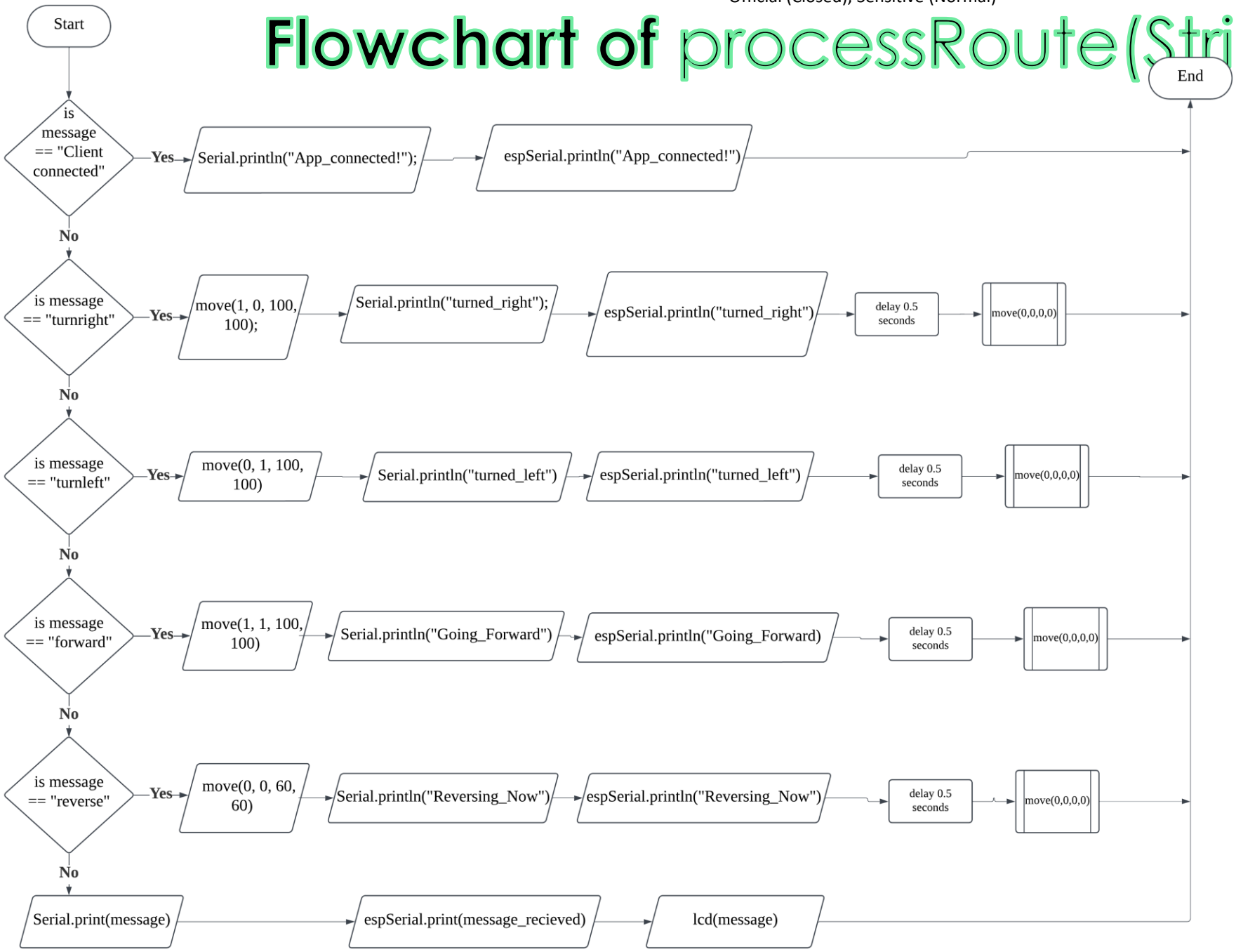
Flowchart of LCD(String message, String message1)



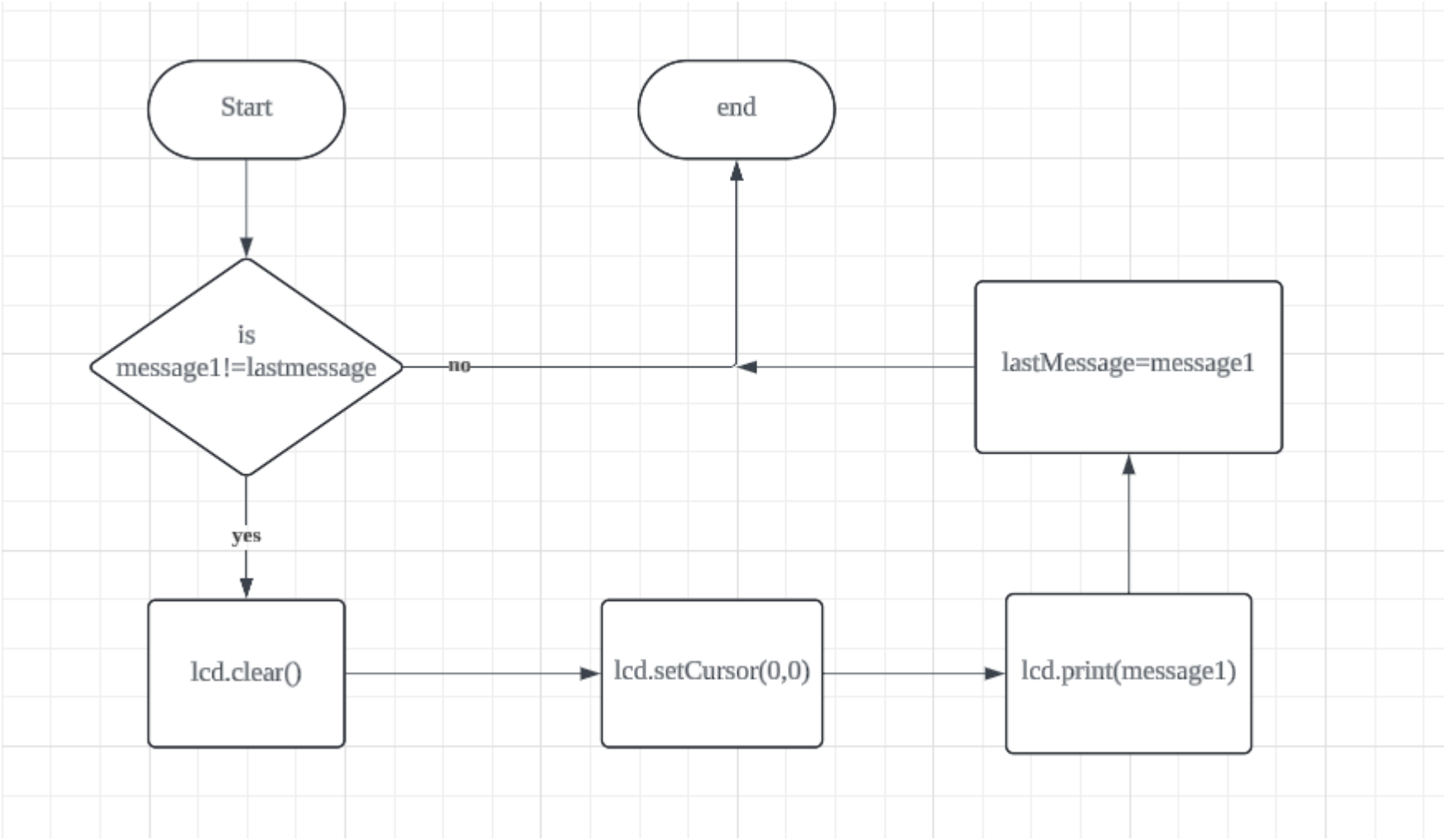
Flowchart of RemoteAccess_UNO_CODE main loop()



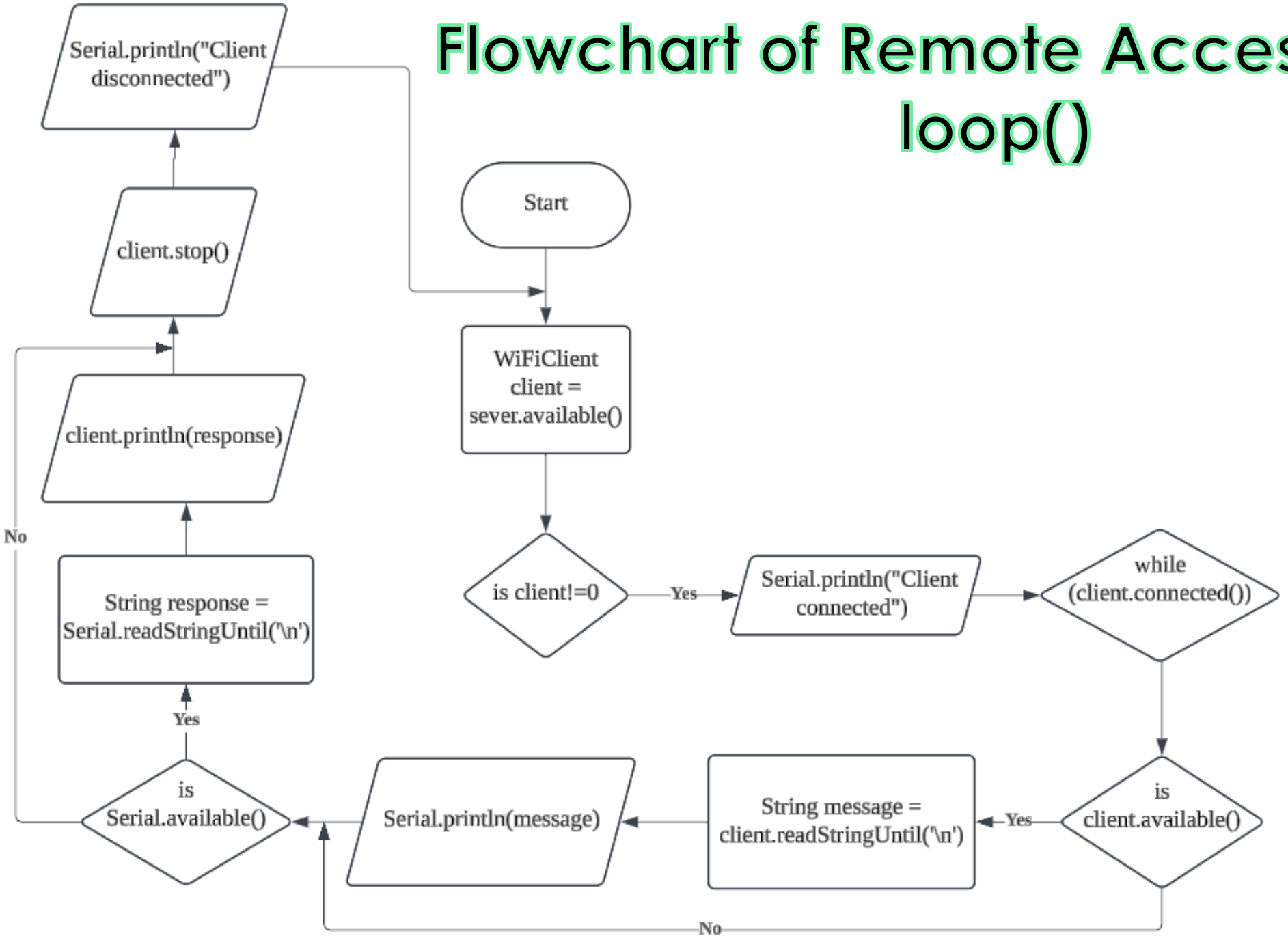
Flowchart of processRoute(String message)



Flowchart of LCD(String message1)



Flowchart of Remote Access ESP-01 loop()



How to run RemoteAccess

Upload ESP-01 Code

- Flash `RemoteAccess_ESP-01_CODE.ino` to ESP-01 using a USB-to-serial adapter.

First-Time WiFi Configuration

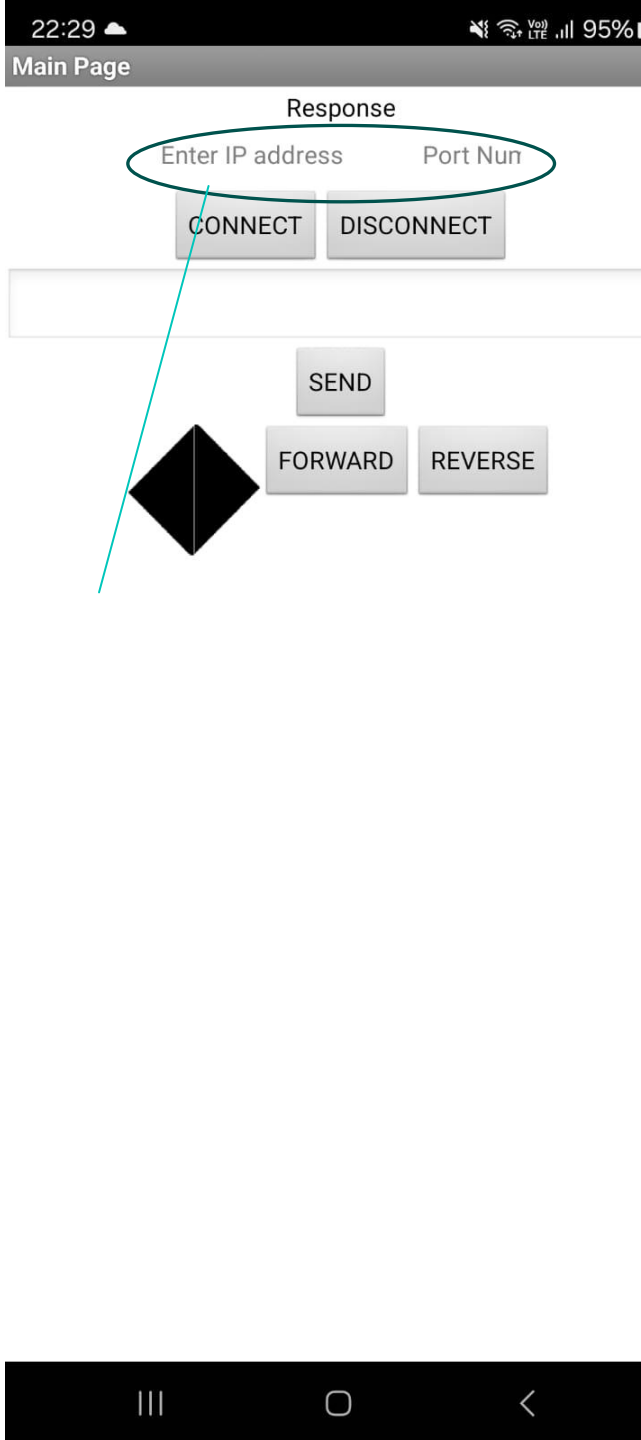
- Connect to the temporary WiFi **"ESP8266_AP"** from a phone/laptop.
- Enter SSID & password of a nearby network.
- Confirm connection via Serial Monitor.
- ESP-01 will auto-connect in future activations.

Upload Uno Code

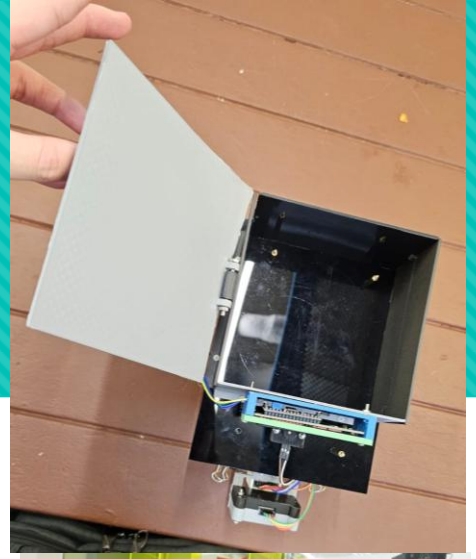
- Flash `RemoteAccess_UNO_CODE.ino` to the Arduino Uno.
- **Retrieve ESP-01 IP & Port**
- Disconnect & reconnect Serial Monitor (**9600 baud**).
- The assigned **IP address & port** will appear.

Connect via Android App

- Install **"App for RemoteAccess" (APK provided).**
- Enter the **IP address & port** shown by the ESP-01.
- If no response:
  - Open MIT App Inventor.
  - Import **IED_WIFIMOD.aia** (provided in submission file).
  - Click **Connect** → **AI Companion** and scan the QR code.
  - Retry connection.
  - **Verify LCD Display Output**
- Press **Connect** in the app.
- Enter text in the input bar → It should display on the Uno's LCD.
- **Test Control Buttons**
- Arrow & Forward/Reverse buttons should function when held down.

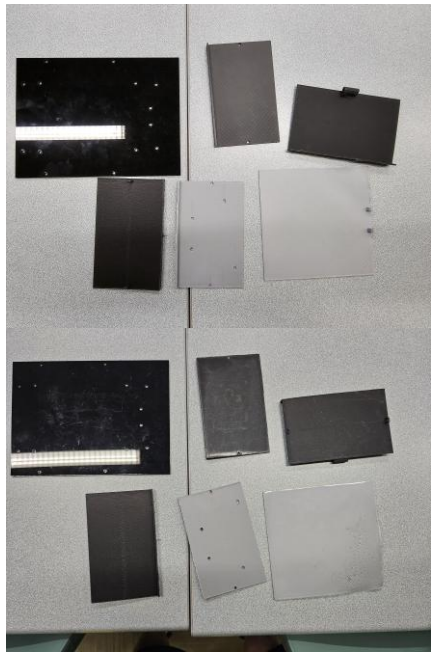


Photos of the final Project

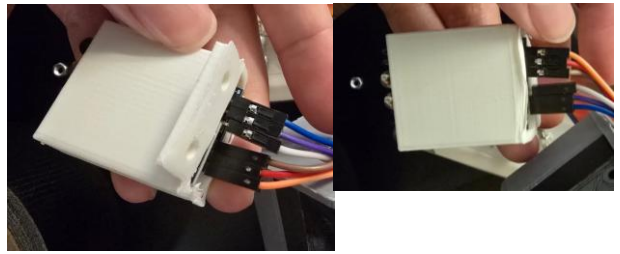


- Show all your 3d Print/laser cut sensors support stand here.

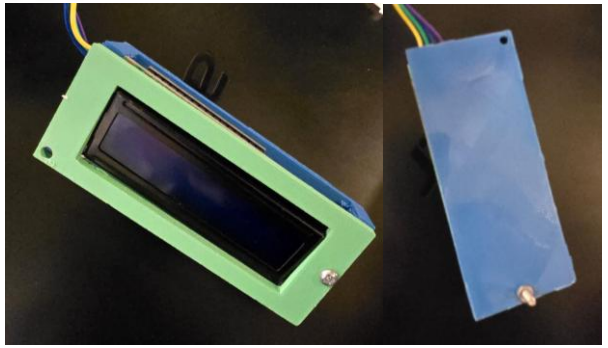
Box walls, cover and base

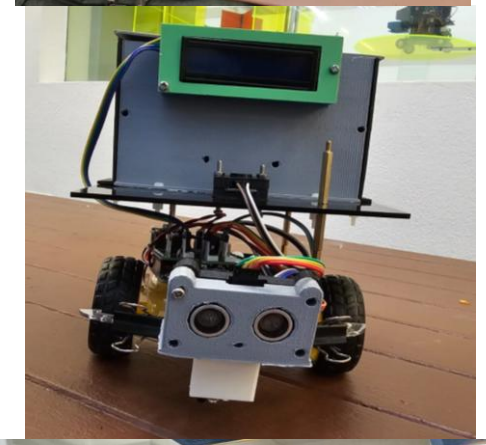


IR sensor

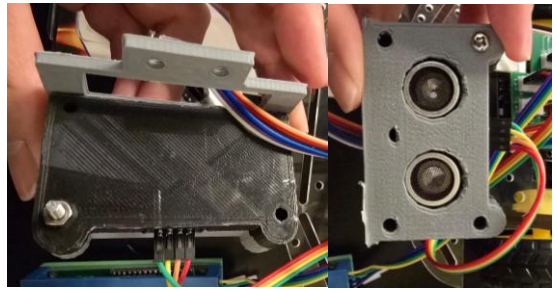


LCD

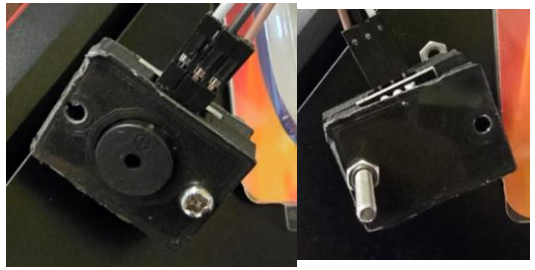




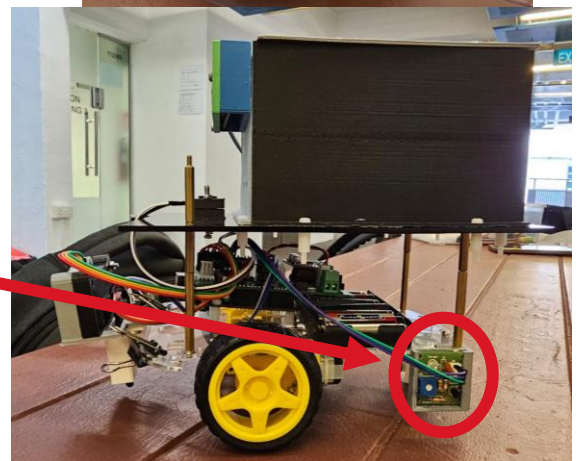
US sensor



Buzzer



Target board



Individual Contribution Including 3d/laser cut, defined Function, if any

S/N	Justin Lui Jun Wei	Brandon Lee Jia Heng	Terence Tng Khai Chun	Nathan Chew Wei Zhi
1	Wrote base code for necessary sensors getdist() Streamlined the direction regarding progress of our group, kept us updated with overall progress	ESP-01 code(TCP server+wifi manager) Uno code for remote access Serial connection for esp01 and uno processRoute()	drivemotor() move() handleInterrupt() updateLCD() playNotes()	LCDstatement() Movementsounds()
2	Debugged remote access and autorun code Bookings and coordinator for 3D prints and team meetings Supplied batteries and nuts and bolts for hardware components	Created remote access application(apk) -Via MIT app inventor 3D-model for target board holder	Integrated updateLCD() and playNotes() with LCDstatement() and Movementsounds() Integrated LCDstatement() and Movementsounds() with drivemotor() Programmed interrupt for IR and target board	Integrated LCDstatement() and Movementsounds() with drivemotor() and getdist() Helped with Remoteaccess UNO code and provided necessary materials like bolts, mobile hotspot
3	Generated ideas for how target board can be used, helped find ways to simplify motor code	Integrating : processRoute() LCD() For uno code(remote control)	3D-models for IR and US holders and box lid laser cut for box base	3D-models for LCD,buzzer holders and box walls
4	sketch slide Poster	remoteaccess flowchart	drivemotor flowchart Poster	All other Wifi and sensor flowcharts Poster

References